Should Platforms Be Allowed to Sell on Their Own Marketplaces?

RAND Journal of Economics, 2022
Group presentation (40 minutes)
May 18, 2026

Author: Andrei Hagiu, Tat-How Teh, and Julian Wright

Reporters: Jinghao Mu, Jialong Zheng, Binghui Guo, Zhiyuan Chen, Zhuoyue Wu

Lingnan College, Sun Yat-Sen University

Contents

01

PART 1

Research Question

Presenter: Jinghao Mu

Motivation: A Dual Role platform

Many large platforms now play two roles at the same time.
(Eg. Amazon, JD.com, Apple App Store...)

- They run a marketplace for third-party sellers.
- They also sell their own products on the same marketplace.

The key tension

The platform is both the market **organizer** and one of the **sellers** in the market.

Why Dual Mode Is Controversial

Dual mode has both potential harms and potential efficiency benefits.

Regulatory concerns

- **Product imitation:** copy successful sellers using marketplace data.
- **Self-preferring:** steer consumers toward the platform's own products.
- **Innovation harm:** weaken third-party sellers' investment incentives.

Potential benefits

- **One-stop shopping:** consumers can compare more options in one place.
- **Efficient supply:** the platform may be a better seller in some categories.
- **Price pressure:** the platform's product can constrain third-party prices.

Policy question

Should regulators ban dual mode itself, or target specific harmful conduct?

Research Question

The paper asks three linked questions.

1. Does **banning** dual mode always **raise consumer surplus and welfare**?
2. If dual mode is banned, does the platform **switch to seller mode or marketplace mode**?
3. Are targeted **rules on imitation and self-preferring** better than a full ban?

Key Insight

The problem is not dual mode itself, but how the platform uses its power.

- A ban on dual mode is not always good for consumers or welfare.
- Rules that target imitation and self-preferring can work better.

Location and Contribution

The paper is located at the intersection of platform economics and antitrust policy.

- It studies a platform that can choose among **marketplace mode**, **seller mode**, and **dual mode**.
- It connects the policy debate on platform bans with a formal welfare analysis.
- It separates **dual mode itself** from two possible bad conducts: **product imitation** and **self-preferring**.

Presentation Roadmap

The presentation is organized into five parts.

1. Part 1: Motivation and main question	Jinghao Mu
2. Part 2: Baseline model setup	Jialong Zheng
3. Part 3: Baseline equilibria and price squeeze	Binghui Guo
4. Part 4: Ban on dual mode	Zhiyuan Chen
5. Part 5: Imitation, self-preferring, and policy comparison	Zhuoyue Wu

02

PART 2

Baseline Model

Presenter: Jialong Zheng

Players I: Consumers and Competitive Benchmark

Consumers

- Outside option of not buying anything (e.g., postpone purchase or buy another product): $v_o \sim G$.[a]
- Platform transaction benefit: $b > 0$ (finding products, one-click payment, delivery tracking, easy returns).
- Choice rule: buy the option with the highest net utility.

[a] G is log-concave and continuously differentiable on $[0, \bar{v}_o]$, with density g.

Fringe sellers

- At least two identical sellers.
- Sell a basic product with value v.
- Consumers already know them and can buy directly.
- Bertrand competition gives the price benchmark.[a]

[a] All marginal costs are normalized to zero.

Players II: Platform and Innovative Seller

Innovative (superior) seller S

- Benefits from innovation; discovered only if listed on M.
- Better product value: $v + \Delta > v > 0$.
- Innovation: chooses quality investment Δ.

Platform M

- Chooses marketplace, seller, or dual mode.
- Seller: sells its own offering, valued at $v + \sigma$.
- Marketplace: sets commission $\tau \geq 0$ for third-party sellers.

Values and Parameters

Product values are:

Product	Consumer value	Meaning
Fringe sellers	v	Basic product
Innovative seller S	$v + \Delta$	Better product
Platform M	$v + \sigma$	Platform's own product

- $b > 0$: convenience benefit from buying through the platform.
- $v_o \sim G$: outside option.
- τ: platform commission.

Intuition

$\sigma > 0$ means the platform is a better seller than fringe sellers; $\sigma < 0$ means the opposite.

Innovation by Seller S

Seller S chooses innovation $\Delta \geq \Delta^l$ and pays fixed cost $K(\Delta)$.

$$K'(\bar{\Delta}) = G(\nu + \sigma + b)$$

Meaning of $\bar{\Delta}$

$\bar{\Delta}$ is the high innovation level that can arise when the market gives strong rewards to innovation.

Intuition

The left side is marginal cost of innovation. The right side is the demand gain from better quality.

Product Discovery

Consumers do not know about S at the start.

- Consumers know the fringe sellers.
- Consumers may know the platform's own product.
- Consumers discover S only if S is available on the platform.

Why this matters

The platform is not only a sales channel. It is also a discovery channel.

Intuition

If a policy changes the platform's mode, it may also change whether consumers find S.

Three Platform Modes

Mode	What M does	Key feature
Marketplace	Hosts third-party sellers	Earns commission τ
Seller	Sells its own product	No marketplace for S
Dual	Hosts sellers and sells own product	Both roles at once

Intuition

The policy debate is about whether the third mode should be allowed.

Timing

The game is solved by backward induction.

1. M chooses its mode and sets commission τ if there is a marketplace.
2. S decides whether to join the platform and chooses Δ.
3. Sellers set prices.
4. Consumers choose what to buy.

Equilibrium concept

Subgame Perfect Nash Equilibrium.

03

PART 3

Equilibrium Analysis

Presenter: Binghui Guo

Advanced Microeconomics Presentation

Marketplace Mode

In marketplace mode, M only charges commission τ.

$$p_i \leq \tau + \Delta, \qquad p_o \leq \tau + \Delta - b$$

Showrooming constraint

If $\tau > b$, seller S can use the platform for discovery and then sell directly.

Intuition

The platform can charge at most the extra convenience value b that it gives consumers.

Marketplace Equilibrium

Proposition 1:

$$\tau^{mkt} = b, \qquad G(\nu) = K'(\Delta^{mkt}), \qquad p_i^* = b + \Delta^{mkt}$$

$$\Pi^{mkt} = bG(\nu), \qquad \pi^{mkt} = \Delta^{mkt}G(\nu) - K(\Delta^{mkt})$$

Intuition

M sets the highest fee that does not push trade outside the platform. S invests until marginal cost equals marginal market gain.

Seller Mode

In seller mode, the platform only sells its own product.

$$\max_{p_m \leq b+\sigma} p_m G(v + b + \sigma - p_m) \tag{3.1}$$

The competitive constraint binds:

$$p_m^* = b + \sigma$$

Proposition 2

M sells to all consumers, while S sells to no one.

Intuition

The platform captures its convenience value and its product advantage, but consumers do not discover S.

Dual Mode: Two Pricing Cases

In dual mode, M both charges commission and sells its own product.

Case 1: $\sigma \geq \Delta$

The platform's own product is good enough. Consumers buy from M.

$$p_i^* = \tau, \qquad p_m^* = \tau + \sigma - \Delta$$

Case 2: $\Delta > \sigma$

Seller S has the better product. But M still puts pressure on S's price.

Intuition

Dual mode matters most when S is better, but M can still discipline S's price.

Price Squeeze

In the price squeeze equilibrium:

$$p_i^* = p_m^* + \Delta - \sigma \tag{3.2}$$

$$p_m^* \in [\max\{\tau - \Delta + \sigma, 0\}, \tau + \min\{\sigma, 0\}]$$

$$\Pi = \tau G(v + \sigma + b - p_m^*)$$

Intuition

M may not sell, but its own product caps S's price. A lower S price raises demand and commission income.

Innovation Constraint

Define $\bar{\tau}$ by:

$$(\bar{\Delta} - \sigma - \bar{\tau})G(\nu + b + \sigma) - K(\bar{\Delta}) = 0$$

Lemma 1:

- If $\tau \leq \bar{\tau}$, then S chooses high innovation: $\Delta = \bar{\Delta}$.
- If $\tau > \bar{\tau}$, then S chooses low innovation: $\Delta = \Delta^l$.

Intuition

A high commission lowers S's margin. If the margin is too low, innovation is not worth the cost.

Dual Mode Equilibrium

Proposition 3: M balances two limits.

- Showrooming constraint: $\tau \leq b$.
- Innovation constraint: $\tau \leq \bar{\tau}$.

If high innovation is profitable enough:

$$\tau^{dual} = \min\{b, \bar{\tau}\}, \qquad \Delta^{dual} = \bar{\Delta}$$

Otherwise:

$$\tau^{dual} = b, \qquad \Delta^{dual} = \Delta^{l}$$

Intuition

The platform may charge less to keep S willing to innovate, because innovation expands the market.

Which Mode Does the Platform Choose?

Corollary 1:

- M prefers marketplace mode over seller mode iff $\sigma \leq 0$.
- Dual mode is always better for M than marketplace mode.
- There is a threshold $\underline{\sigma} > 0$ such that:

$$\Pi^{dual} > \Pi^{sell} \quad \text{iff} \quad \sigma < \underline{\sigma}$$

Intuition

Dual mode creates more transactions through price squeeze. But if M's own product is much better, M may prefer to sell alone.

04

PART 4

Imitation and Self-Preferencing

Presenter: Zhiyuan Chen

Advanced Microeconomics Presentation

What Is Added?

This part adds two practices to the baseline dual-mode model: **product imitation** and **self-preferencing**.

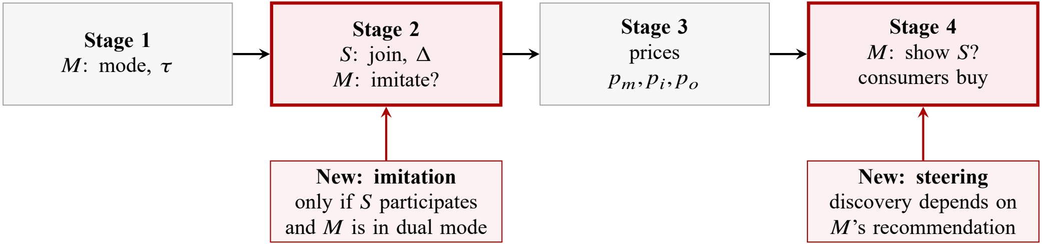
Product imitation

- **Assumption:** copied product gives surplus $v + \Delta$.
- **Choice:** S: participate or not; then M: imitate or not if S participates.
- **Motivation:** marketplace data helps M identify and copy successful innovations.

Self-preferencing

- **Assumption:** consumers discover S through M's recommendation.
- **Choice:** M: recommend S or not after prices are set.
- **Motivation:** consumers often rely on platform search, ranking, or recommendations to find new products.

Timing: Where Section 4 Changes the Game



Changes from the baseline

- M copying choice: M imitates S or not.
- M disclosure choice: M shows S or not.

Review: Exploitative Equilibrium

1. Exploitative equilibrium: M hides S and charges the highest price allowed by outside options.
   - mode: M as seller, don't show S.
2. M exploitative price:

$$p_m^* = \min\{\tau, b\} + \sigma \tag{4.1}$$

   - M vs fringe ($p_f \geq \tau$) through marketplace: $v + \sigma + b - p_m^* \geq v + b - p_f \Rightarrow p_m^* \leq \tau + \sigma$
   - M vs fringe not through marketplace: $v + \sigma + b - p_m^* \geq v \Rightarrow p_m^* \leq b + \sigma$
3. M exploitative profit:

$$\begin{aligned} \Pi^{exploit} &= \Pi^{sell} = p_m^* G(v + b + \sigma - p_m^*) \\ &= (\min\{\tau + b\} + \sigma) G(v + b - \min\{\tau, b\}). \end{aligned} \tag{4.2}$$

Review: Price Squeeze Equilibrium

1. Price-squeeze Equilibrium: M shows S, but caps S's inside price and margin.
   - model: M dual mode, shows S.
2. S Price-squeeze price:

$$p_i^* = \max\{\tau, \Delta - \sigma\} \geq \tau \tag{4.3}$$

   - particpate vs not: $p_i^* - \tau \geq 0 \Rightarrow p_i^* \geq \tau$.
   - sell or not: $\Delta - p_i^* \geq \sigma - p_m^* \Rightarrow p_i^* \leq p_m^* + \Delta - \sigma$. ($M$ sets the lowest price (**as seller**) to squeeze S's)
3. M Price-squeeze price:

$$p_m^* = \max\{\tau - \Delta + \sigma, 0\} \geq 0 \tag{4.4}$$

4. Price-squeeze profits: M get the commission (as marketplace) from S's sales:

$$\begin{aligned} \Pi^{sqz} &= \Pi^{dual} = \tau G(\nu + b + \Delta - p_i^*) = \tau G(\nu + \sigma + b - p_m^*) \\ \pi^{sqz} &= (p_i^* - \tau)G(\nu + b + \Delta - p_i^*) = (p_i^* - \tau)G(\nu + \sigma + b - p_m^*) \end{aligned} \tag{4.5}$$

Consequence of Imitation

Imitation: $\sigma^* = \max\{\Delta, \sigma\}$

If $\sigma \geq \Delta$ no imitation (change), since suppose $\Delta > \sigma \rightarrow \sigma^* = \Delta$

1. Exploitative path: Eq. (4.1), as a seller extracts innovation rents.

$$p_m^* \uparrow = \min\{\tau, b\} + \Delta \tag{4.6}$$

2. Price-squeeze path: Eq. (4.4) and (4.3), as dual mode squeezes prices to generate more sales $p_m^* = p_i^* = \tau$

$$\begin{aligned} p_m^* \downarrow &= \max\{\tau, \Delta - \Delta\} = \tau \\ p_i^* \downarrow &= \max\{\tau + \Delta - \Delta, 0\} = \tau \end{aligned} \tag{4.7}$$

3. S's innovation: Eq. (4.5), no innovation $\Delta^* = \Delta^l$

$$\Delta^* \downarrow = \arg\max_{\Delta \geq \Delta^l} \{-K(\Delta)\} = \Delta^l \tag{4.8}$$

Consequence of Self-Preferencing

Self-preferencing: $\tau \leq b \rightarrow \tau \leq b + \Delta$

If

1. Recommendation rule:
$$\Delta + b - p_i \geq \max\{\Delta - p_o,\ b - \tau,\ 0\} \tag{4.9}$$
2. High-fee path: M can extract S's margin as a marketplace.
$$\tau^* = b + \Delta, \qquad p_i^* = \tau^*, \qquad \pi_S = -K(\Delta) \tag{4.10}$$
3. Hiding path: M shields its own product from competition with S.
$$p_m^* = \min\{\tau, b\} + \sigma \tag{4.11}$$

Consequence of Self-Preferencing

Self-preferencing means that discovery of S is no longer automatic: after observing prices, M can show S or hide it.

Showrooming constraint is relaxed

Without steering, S can use the platform for discovery and then direct consumers to its outside channel, so

$$\tau \leq b. \tag{4.12}$$

With steering, a high-fee price squeeze can still be sustained:

$$\tau = b + \Delta, \qquad p_i^* = \tau, \qquad \Pi_M = (b + \Delta)G(\nu), \qquad \pi_S = -K(\Delta). \tag{4.13}$$

Competition from S can also be switched off

If M hides S, consumers compare M only with fringe sellers and no purchase. With imitation, this gives

$$p_m^* = b + \Delta, \qquad \Pi_M = (b + \Delta)G(\nu), \qquad CS = CS^0. \tag{4.14}$$

Economic meaning: self-preferencing removes the visibility threat that used to discipline M's commission and pricing. The platform can either tax S through a high fee or steer consumers to its own product, so on-platform price competition largely disappears.

Marketplace and Seller Mode

Marketplace mode: same commission τ from any marketplace sale.

If M shows S

Buy S through the marketplace iff:

$$\Delta + b - p_i \geq \max\{\Delta - p_o,\ b - \tau,\ 0\} \tag{4.15}$$

- Left: surplus from S on M.
- Right: other best outside option.

If M hides S

Buy a fringe seller on the marketplace iff:

$$b - \tau \geq 0 \tag{4.16}$$

- Left: surplus from a fringe seller on M.
- Right: no purchase.

Recommendation rule

- Show S if Eq. (4.15), holds: M earns commission τ.
- Hide S if Eq. (4.15), fails but Eq. (4.16), holds.

Seller mode is unchanged: M sells only its own product.

Proposition 5: Dual Mode Equilibrium

Assumptions in the regulated-practice environment

- **Imitation allow:** M can perfectly imitate S's innovation at no cost, $\sigma = \max\{\Delta, \sigma\}$.
- **Self-preferencing allow:** M can perfectly steer consumers toward its own offer, $\tau = b + \Delta$.
- **Dual mode is feasible:** M both sells its own product and hosts S.

Main variables in equilibrium

- M's profit: $\Pi^{dual} = \max\left\{p_m^* G(v + b + \Delta^{dual} - p_m^*), \tau^* G(v + b + \Delta^{dual} - p_i^*)\right\} = (b + \max\{\sigma, \Delta^l\})G(v)$.
- S's operating profit: $\pi^{dual} = \max\left\{(p_i^* - \tau)G(v + b + \Delta^{dual} - p_i^*) - K(\Delta^{dual}), 0\right\} = 0$.
- Innovation level: $\Delta^{dual} = \Delta^l \quad \leftarrow \quad \Delta^{dual} = \arg\max_{\Delta^{dual} \geq \Delta^l} \pi^{dual} - K(\Delta)$.
- Consumer surplus: $CS^{dual} = CS^0 \equiv \int_0^\infty \max\{v, v_o\}\, dG(v_o)$ ≡f(v).
- Total welfare: $W^{dual} = \Pi^{dual} + \pi^{dual} + CS^{dual}$

Proposition 5: Outcome Vector

Table 1: Dual-mode benchmark with imitation and self-preferencing (Proposition 5)

Case	M's choice of mode	τ	Π	π	Δ
$\sigma \geq \Delta^l$	Dual - no imitation, M sells	$\tau^* \geq b + \Delta^l$	$(b+\sigma)G(\nu)$	0	Δ^l
$0 < \sigma < \Delta^l$	Dual - high fee or imitation	$\tau^* \geq b$	$(b+\Delta^l)G(\nu)$	0	Δ^l
$\sigma \leq 0$	Dual - high fee or imitation	$\tau^* \geq b$	$(b+\Delta^l)G(\nu)$	0	Δ^l

High-fee and imitation paths

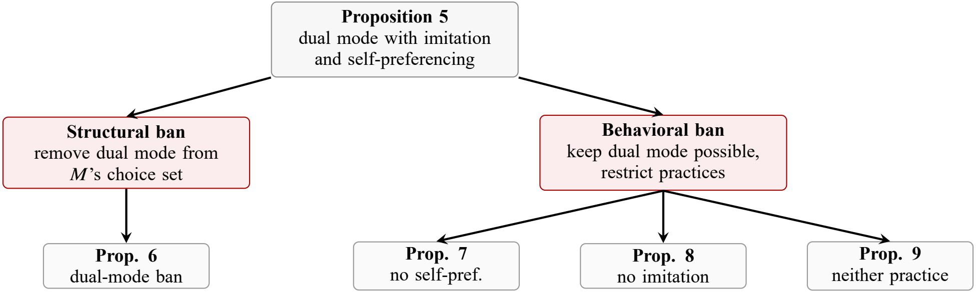
- **High fee** (M as marketplace): $\tau = b + \Delta^l$, from self-preferencing M may choose to not show S, thereby capturing S's full profit.
- **Imitation** (M as seller): $p_m = b + \Delta^l$.

- $\pi = 0$: (1) with a high fee, M extracts all of S's price margin; (2) with imitation, M makes all sales itself.
- $\Delta = \Delta^l$: since S's operating profit does not increase with Δ, it has no incentive to innovate above the minimum level.

Policy Map: How the Bans Are Related

Ban types

- **Structural ban**: change M's business model by removing dual mode; it can shift M to a pure seller or pure marketplace mode.
- **Behavioral ban**: keep dual mode possible but remove specific practices; it can restore price discipline and/or innovation incentives.



Dual-Mode Ban - Setup

Dual-Mode Ban

- **Economic meaning**: M can only choose between being a **seller** or a **marketplace**, but can still choose **imitation** or **self-preference** behavior.
- **Mathematical meaning**: M's stage-0 choice set is reduced from {Marketplace, Seller, Dual} to {Marketplace, Seller}.

Mode	τ^*	Δ	Π	π	CS
Seller	$\tau^* \geq b$	$\Delta^* = \Delta^l$	$(b+\sigma)G(v)$	0	CS^0
Marketplace	$\tau^* \leq b$	$\Delta^* = \Delta^{mkt}$	$\tau^* G(v)$	$\Delta^{mkt} G(v) - K(\Delta^{mkt})$	CS^0

- $p_m^* = \min\{b, \tau^*\} + \sigma$ and $\tau^* = b$, see Eq. (**??**).
- $p_i^* = \min\{\Delta^*, b + \Delta^* - \tau^*\}$, see Eq. (**??**).

Dual-Mode Ban - Equilibrium

Table 2: Effects of a dual-mode ban (Proposition 6) v.s. Table 1

Region	M's choice of mode	Π	π	$CS = CS^0$	Δ	W
$\sigma \geq \Delta^l$	Seller	[=]	0 [=]	[=]	Δ^l [=]	[=]
$0 < \sigma < \Delta^l$	Seller	$(b+\sigma)G(\nu)$ [↓]	0 [=]	[=]	Δ^l [=]	$(b+\sigma)G(\nu) + CS^0$ [↓]
$\sigma \leq 0$	Marketplace	$bG(\nu)$ [↓]	$\Delta^{mkt}G(\nu) - K(\Delta^{mkt})$ [↑]	[=]	Δ^{mkt} [↑]	$(b+\Delta^{mkt})G(\nu) - K(\Delta^{mkt}) + CS^0$ [↑]

- $0 < \sigma < \Delta^l$: **(1) math**: $(b+\sigma)G(\nu) < (b+\Delta^l)G(\nu) \rightarrow W \downarrow$. **(2) econ**: M to switch to seller mode, so S's superior product is no longer discovered through the platform.
- $\sigma \leq 0$: **(1) math**: $\cancel{bG(\nu)} + \Delta^{mkt}G(\nu) - K(\Delta^{mkt}) > \cancel{bG(\nu)} + \Delta^l G(\nu) \rightarrow W \uparrow$. **(2) econ**: M to switch to marketplace mode, so S's inferior product is now discovered through the platform.

中山大學嶺南(大學)學院 LINGNAN (UNIVERSITY) COLLEGE

Self-Preferencing Ban - Setup

Self-Preferencing Ban

- **Economic meaning**: if S participates M cannot steer consumers away from S.
- **Mathematical meaning**: the disclosure choice is removed and the showrooming constraint is restored ($\tau^* \leq b$).

- For M: $\tau^* = b, p_m^* = b + \max\{\sigma - \Delta^*, 0\}$ (no imitation, imitation) and $\Pi = \max\left\{\tau^* G(\nu + b + \Delta^l - p_i^*), p_m^* G(\nu + b + \sigma - p_m^*)\right\} = (b + \max\{\sigma - \Delta^l, 0\}) G(\nu + \Delta^*)$ (seller, marketplace).
- For S: $\Delta = \Delta^l, p_i^* = b, p_o^* = 0$ and $\pi = \max\left\{(p_i^* - \tau^*) G(\nu + b + \Delta^* - p_i^*), p_o^* G(\nu + \Delta^* - p_o^*)\right\} = 0$ (through M, directly).

Regime	Outcome	τ	Δ	p_m^*	p_s^*	Π	π
$\sigma < \Delta^l$	Dual, imitation, S sells	b	Δ^l	b	$p_i^* = b$	$bG(\nu + \Delta^l)$	0
$\sigma \geq \Delta^l$	Dual, no imitation, M sells	b	Δ^l	$b + \sigma - \Delta^l$	$p_i^* = b$	$(b + \sigma - \Delta^l) G(\nu + \Delta^l)$	0

Self-Preferencing Ban - Equilibrium

Table 3: Effects of banning self-preferencing (Proposition 7) v.s. Table 1

Region	M's choice of mode	Π	π	CS	Δ	W
$\sigma \geq \Delta^l$	Seller	[=]	[=]	[=]	[=]	[=]
$\underline{\sigma}^{steer} < \sigma < \Delta^l$	Seller	$(b+\sigma)G(\nu)$ [↓]	[=]	[=]	[=]	[↓]
$\sigma \leq \underline{\sigma}^{steer}$	Dual, no self-pref.	$bG(\nu+\Delta^l)$ [↓]	[=]	CS^{ns} [↑]	[=]	$\Pi + CS^{ns}$ [↑]

Note: $CS^{ns} \equiv \int_0^\infty \max\{\nu+\Delta^l, \nu_o\}\, dG(\nu_o)$; $CS^0 \equiv \int_0^\infty \max\{\nu, \nu_o\}\, dG(\nu_o)$.

- $\underline{\sigma}^{steer} < \sigma < \Delta^l$: **math**: $(b+\sigma)G(\nu) < (b+\Delta^l)G(\nu)$. **econ**: M switches to seller mode; S is not discovered.
- $\sigma \leq \underline{\sigma}^{steer}$:
  - $\Pi \downarrow$: **math**: $bG(\nu+\Delta^l) < (b+\Delta^l)G(\nu)$ by assumption 2. **econ**: restored competition lowers prices and expands demand.
  - $W \uparrow$: **math**: $bG(\nu+\Delta^l) + CS^{ns} > (b+\Delta^l)G(\nu) + CS^0 \Leftrightarrow b\left[G(\nu+\Delta^l) - G(\nu)\right] + \int_\nu^{\nu+\Delta^l} (\nu+\Delta^l-\nu_o)\, dG(\nu_o) > 0$. **econ**: consumers are better off with S's product being discovered, and get the more consumer.

Imitation Ban - Setup

Imitation Ban

- **Economic meaning**: M can credibly commit not to copy S's product, S again benefit from innovation.
- **Mathematical meaning**: M can only effect high innovation by choosing a fee $(\tau \leq \bar{\tau})$ [a]

[a] $\bar{\tau}$ is the highest fee that allows S to have high innovation, satisfied $(\bar{\Delta} - \sigma - \bar{\tau})G(\nu + b + \sigma) - K(\bar{\Delta}) = 0$.

Innovation condition (Eq. (4.17) or Eq. (4.18) holds)

- Innovation-preserving condition for M:

$$\bar{\tau}G(\nu + \sigma + b) \geq (b + \max\{\sigma, \Delta^l\})G(\nu). \tag{4.17}$$

left - M's profit from preserving innovation; right - M's profit from from rent extraction.

- Non-binding innovation constraint for S:

$$b + \Delta^l \leq \bar{\tau}. \tag{4.18}$$

left - the extractive fee; right - the highest fee compatible with high innovation.

Imitation Ban - Equilibrium

Table 4: Effects of banning imitation (Proposition 8) v.s. Table 1

Region	M's mode	Π	π	CS	Δ	W
Innovation condition fails	Dual, no imitation	$(b+\max\{\sigma,\Delta^l\})G(\nu)$ [=]	0 [=]	CS^0 [=]	Δ^l [=]	$\Pi + CS^0$ [=]
Innovation condition holds	Dual, no imitation	$\bar{\tau}G(\nu+\sigma+b)$ [↑]	0 [=]	CS^{ni} [↑]	$\bar{\Delta}$ [↑]	$\Pi + CS^{ni}$ [↑]

$CS^{ni} \equiv \int_0^\infty \max\{\nu+b+\sigma, \nu_o\}\, dG(\nu_o)$. $CS^0 \equiv \int_0^\infty \max\{\nu, \nu_o\}\, dG(\nu_o)$.

- Condition fails: **math**: $\tau = b + \Delta^l$ and $\Delta = \Delta^l$, so the outcome equals Proposition 5. **econ**: innovation condition fails.
- Condition holds:
  - $\Pi \uparrow$: **math**: $\bar{\tau}G(\nu+\sigma+b) > (b+\Delta^l)G(\nu)$, by Eq (4.17) or Eq (4.18). **econ**: commitment not to copy lets M induce high innovation and sell to a larger market.
  - $W \uparrow$: **math**: $\Pi \uparrow, CS \uparrow$. **econ**: higher innovation expands demand and consumers benefit.

Ban Both Practices - Setup

Ban Imitation and Self-Preferencing

- **Economic meaning**: M compete on price and quality, but can still choose to be a seller or a marketplace.
- **Mathematical meaning**: $\Delta = \bar{\Delta}$ (no imitation) and $\tau \leq b$ (no self-preferencing).

Table 5: Post-ban mode logic

Mode selected by M	Key constraint	Innovation	Main mechanism
Seller	no marketplace for S	Δ^l	discovery of S is lost
Dual	$\tau \leq b$ and possibly $\tau \leq \bar{\tau}$	Δ^l or $\bar{\Delta}$	price squeeze plus innovation incentive

Ban Both Practices - Equilibrium

Table 6: Effects of banning both practices (Proposition 9) v.s. Table 1

Region	M's choice of mode	Π	π	CS	Δ	W
$\sigma \geq \max\{\underline{\sigma}, \Delta^l\}$	Seller	[=]	[=]	[=]	[=]	[=]
$\underline{\sigma} < \sigma < \Delta^l$	Seller	[↓]	[=]	[=]	[=]	[↓]
$\sigma \leq \underline{\sigma}$	Dual, no imitation or self-pref.	Π^{P3} [↓]	π^{P3} [↑]† or [=]	CS^{P3} [↑]	Δ^{P3} [↑]‡ or [=]	$\Pi^{P3} + \pi^{P3} + CS^{P3}$ [↑]

$(\Pi^{P3}, \pi^{P3}, CS^{P3}, \Delta^{P3})$: baseline dual-mode outcome in Proposition 3.
†: π increases if $b < \bar{\tau}$; ‡: Δ increases if $b \leq \bar{\tau}$ or condition (8) holds.

- $\underline{\sigma} < \sigma < \Delta^l$: **math**: $(b + \sigma)G(\nu) < (b + \Delta^l)G(\nu)$, so $W \downarrow$. **econ**: M switches to seller mode and S is no longer discovered.
- $\sigma \leq \underline{\sigma}$: **math**: $W^{P3} > (b + \Delta^l)G(\nu) + CS^0$. **econ**: the policy restores both visibility and innovation incentives while preserving dual-mode competition.

Policy Comparison: $\sigma > 0$

Table 7: Behavioral remedies relative to a dual-mode ban (Corollary 2)

Region	Remedy	Π	π	CS	Δ	W
$\sigma \in (0, \underline{\sigma}^{steer}]$	Ban self-preferencing	$\uparrow$	$=$	$\uparrow$	$=$	$\uparrow$
$\sigma > \underline{\sigma}^{steer}$	Ban self-preferencing	$=$	$=$	$=$	$=$	$=$
All $\sigma > 0$	Ban imitation	$\uparrow$	$=$	$\uparrow$	$\uparrow$	$\uparrow$
$\sigma \in (0, \underline{\sigma}]$	Ban both practices	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$
$\sigma > \underline{\sigma}$	Ban both practices	$=$	$=$	$=$	$=$	$=$

Baseline: under a dual-mode ban and $\sigma > 0$, M chooses seller mode. Arrows are weak changes.

Policy Comparison: $\sigma \leq 0$

Table 8: Behavioral remedies relative to a dual-mode ban (Corollary 2)

Remedy	Π	π	CS	Δ	W
Ban self-preferencing	$\uparrow$	$\downarrow$	$\uparrow$	$\downarrow$	$\uparrow$ if condition (9) holds $\downarrow$ otherwise
Ban imitation	$\uparrow$	$\downarrow$	$\uparrow$	$\uparrow$ if Eq. (4.17) holds $\downarrow$ otherwise	$\uparrow$ if Δ increases $\downarrow$ otherwise
Ban both practices	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$ if $b \leq \bar{\tau}$ or condition (8) holds $\downarrow$ otherwise	$\uparrow$ if Δ increases or condition (9) holds $\downarrow$ otherwise

Section 4 Takeaways

1. Perfect imitation and perfect steering make dual mode weakly best for M.
2. In dual mode, S chooses Δ^l because innovation rents are extracted.
3. A dual-mode ban does not directly fix copying or steering.
4. Banning self-preferencing restores price competition but not innovation.
5. Banning imitation restores innovation when commitment is valuable enough.
6. Banning both practices restores the baseline dual-mode mechanisms.

Main policy message

The harmful conduct is often a better regulatory target than the platform's dual role itself.

05

PART 5

Policy and Extensions

Presenter: Zhuoyue Wu

Advanced Microeconomics Presentation